

PROGRAMMING THE BASIC COMPUTER

- Introduction
- Machine Language
- Assembly Language
- Assembler
- Program Loops
- Programming Arithmetic and Logic Operations
- Subroutines
- Input-Output Programming



INTRODUCTION

Those concerned with computer architecture should have a knowledge of both hardware and software because the two branches influence each other.

Instruction Set of the *Basic Computer*

Symbol	Hexa code	Description
AND	0 or 8	AND M to AC
ADD	1 or 9	Add M to AC, carry to E
LDA	2 or A	Load AC from M
STA	3 or B	Store AC in M
BUN	4 or C	Branch unconditionally to m
BSA	5 or D	Save return address in m and branch to m+1
ISZ	6 or E	Increment M and skip if zero
CLA	7800	Clear AC
CLE	7400	Clear E
CMA	7200	Complement AC
CME	7100	Complement E
CIR	7080	Circulate right E and AC
CIL	7040	Circulate left E and AC
INC	7020	Increment AC, carry to E
SPA	7010	Skip if AC is positive
SNA	7008	Skip if AC is negative
SZA	7004	Skip if AC is zero
SZE	7002	Skip if E is zero
HLT	7001	Halt computer
INP	F800	Input information and clear flag
OUT	F400	Output information and clear flag
SKI	F200	Skip if input flag is on
SKO	F100	Skip if output flag is on
ION	F080	Turn interrupt on
IOF	F040	Turn interrupt off

m: effective address

M: memory word (operand)
found at m



MACHINE LANGUAGE

- Program

A list of instructions or statements for directing the computer to perform a required data processing task

- Various types of programming languages

- Hierarchy of programming languages

- Machine-language

- Binary code

- Octal or hexadecimal code

- Assembly-language

(Assembler)

- Symbolic code

- High-level language

(Compiler)



COMPARISON OF PROGRAMMING LANGUAGES

• Binary Program to Add Two Numbers

Location	Instruction Code
0	0010 0000 0000 0100
1	0001 0000 0000 0101
10	0011 0000 0000 0110
11	0111 0000 0000 0001
100	0000 0000 0101 0011
101	1111 1111 1110 1001
110	0000 0000 0000 0000

• Hex program

Location	Instruction
000	2004
001	1005
002	3006
003	7001
004	0053
005	FFE9
006	0000

• Program with Symbolic OP-Code

Location	Instruction	Comments
000	LDA 004	Load 1st operand into AC
001	ADD 005	Add 2nd operand to AC
002	STA 006	Store sum in location 006
003	HLT	Halt computer
004	0053	1st operand
005	FFE9	2nd operand (negative)
006	0000	Store sum here

• Assembly-Language Program

ORG	0	/Origin of program is location 0
LDA	A	/Load operand from location A
ADD	B	/Add operand from location B
STA	C	/Store sum in location C
HLT		/Halt computer
A, DEC	83	/Decimal operand
B, DEC	-23	/Decimal operand
C, DEC	0	/Sum stored in location C
END		/End of symbolic program

• Fortran Program

```

INTEGER A, B, C
DATA A,83 B,-23
C = A + B
END

```

